

Files by Study Priority

Every file in the project, ordered **highest** → **lowest priority** for the oral exam.

Legend: 🔴 must know cold · 🟠 know well · 🟡 know roughly · ⚪ just know it exists  = new work for the exam · ♻️ = carried/adapted from Oblig 1/2/3

⚠️ **Caveat:** NOTES.md says *all* of Oblig 1/2/3 was used as starter code, so /♻️ is my best inference from the README's "new features" list + code comments — correct it from memory.

Strategy: study top-down. Learn ONE resource end-to-end (route → validator → controller → service → model) and the rest follow the same shape. Spend real time on the 🔴 block.

🔴 TIER 1 — Must know cold (hardest + most-asked + new)

1. `backend/project/webSockets/gameSocket.js` —  the real-time game engine. The single most complex + most-asked file (rolling, betting, timers, scoring, rejoin).
 2. `frontend/src/pages/Game.jsx` —  game UI + WebSocket client; opens the socket, handles phases, cleanup on unmount.
 3. `frontend/src/components/dice-poker-board.js` — ♻️ (ported from oblig1) Web Component board. Exam *requires* Web Components — know lifecycle + `CustomEvent`.
 4. `frontend/src/components/dice-poker-die.js` — ♻️ single die Web Component (`observedAttributes`, `attributeChangedCallback`).
 5. `backend/project/utils/handEvaluator.js` — ♻️ (ported from oblig1) hand ranking + `calculateEloDeltas` (pairwise ELO).
 6. `backend/project/utils/jwt.js` —  access/refresh token create + verify.
 7. `backend/project/utils/hash.js` —  bcrypt password hashing + salt.
 8. `backend/project/middleware/role.js` —  decode JWT, IP-mismatch check, `requireUser` / `requireAdmin`.
 9. `frontend/src/contexts/AuthContext.jsx` —  login state, `sessionStorage` restore, 30s ban poll.
 10. `frontend/src/api/config.js` —  `fetchWithAuth` + auto refresh-on-401 + `handleResponse`.
 11. `backend/project/services/user.service.js` —  auth logic: register, login, verify email, reset, refresh, ban.
 12. `backend/project/server.js` —  request flow, CORS, rate limiter + security-incident logging, route wiring.
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🟠 TIER 2 — Know well (the rest of the new exam features)

Tournaments  13. `backend/project/services/tournament.service.js` — `startNextRound` random pairing, standings. 14. `backend/project/models/Tournament.js` 15. `backend/project/controllers/tournament.controller.js` 16.

backend/project/routes/tournament.routes.js 17.
backend/project/validators/tournament.validator.js 18.
frontend/src/pages/TournamentPage.jsx 19. frontend/src/pages/Tournament.jsx 20.
frontend/src/pages/admin/TournamentCreate.jsx 21.
frontend/src/pages/admin/TournamentEdit.jsx 22. frontend/src/api/tournaments.js 23.
frontend/src/components/TournamentCard.jsx

Match / game backend (coins + wager are new) 24.

backend/project/services/match.service.js — recordMatch (ELO, coin payout, wager lock). 25.
backend/project/models/Match.js 26. backend/project/controllers/match.controller.js 27.
backend/project/routes/match.routes.js 28. backend/project/validators/match.validator.js
29. frontend/src/api/matches.js 30. frontend/src/components/BettingControls.jsx — 
fold/bet/raise/match controls. 31. frontend/src/components/PlayerInfo.jsx —  per-player
hand/stack/status.

Points / coins  32. backend/project/utils/coins.js — weekly 100-coin grant logic. 33.
backend/project/services/scheduler.js — grantWeeklyCoinsBatch on startup.

Admin + security + activity  34. backend/project/models/Security.js — incident log (rate-limit, IP-
mismatch). 35. backend/project/services/admin.service.js 36.
backend/project/controllers/admin.controller.js 37.
backend/project/routes/admin.routes.js 38. backend/project/services/activity.service.js
39. backend/project/controllers/activity.controller.js 40.
backend/project/routes/activity.routes.js 41. frontend/src/pages/admin/Dashboard.jsx 42.
frontend/src/pages/admin/Users.jsx 43. frontend/src/pages/admin/Comments.jsx 44.
frontend/src/pages/admin/AdminLayout.jsx 45. frontend/src/components/AdminRoute.jsx 46.
frontend/src/components/BanModal.jsx 47. frontend/src/api/admin.js 48.
frontend/src/api/adminUsers.js 49. frontend/src/api/activity.js

Comments (real-time via WebSocket)  50. backend/project/models/Comment.js 51.
backend/project/services/comment.service.js 52.
backend/project/controllers/comment.controller.js 53.
backend/project/routes/comment.routes.js 54.
backend/project/validators/comment.validator.js 55. frontend/src/api/comments.js 56.
frontend/src/components/CommentList.jsx 57. frontend/src/components/CommentItem.jsx 58.
frontend/src/components/CommentForm.jsx

Trophies  59. backend/project/models/Trophy.js 60.
backend/project/services/trophy.service.js 61.
backend/project/controllers/trophy.controller.js 62.
backend/project/routes/trophy.routes.js 63.
backend/project/validators/trophy.validator.js 64. frontend/src/api/trophies.js 65.
frontend/src/components/TrophyBadge.jsx

Email + new auth pages  66. backend/project/utils/mailer.js — verification + reset emails
(Ethereal SMTP). 67. frontend/src/pages/VerifyEmail.jsx 68.

frontend/src/pages/ResetPassword.jsx 69. frontend/src/pages/Error404.jsx —  not-found page (new requirement).

Custom hooks (mostly new, support live updates) 70. frontend/src/hooks/usePolling.js —  interval + AbortController cleanup. 71. frontend/src/hooks/useFetch.js —  generic fetch with cancel. 72. frontend/src/hooks/useDebouncedValue.js —  search debounce. 73. frontend/src/hooks/useSoundEffects.js — sound playback. 74. frontend/src/hooks/useLobbyGames.js — filterLobbyMatches (a plain helper, NOT a hook).

Shared backend infrastructure 75. backend/project/middleware/error.js — central error handler. 76. backend/project/middleware/upload.js — Multer in-memory image upload. 77. backend/project/utils/customError.js — CustomError (status + code). 78. backend/project/validators/validate.js — runs validators, returns 400. 79. backend/project/config/constants.js — limits, enums, coin amounts. 80. backend/project/config/db.config.js — connectDB / disconnectDB. 81. frontend/src/main.jsx — mounts app, wraps in providers. 82. frontend/src/App.jsx — routes, lazy loading, AdminRoute guard.

TIER 3 — Know roughly (Oblig 3 baseline, extended for exam)

Game categories (18 fixed variants)  83. backend/project/models/GameCategory.js 84. backend/project/services/gameCategory.service.js 85. backend/project/controllers/gameCategory.controller.js 86. backend/project/routes/gameCategory.routes.js 87. backend/project/validators/gameCategory.validator.js 88. frontend/src/api/gameCategories.js

User (base CRUD; auth bits live in user.service above)  89. backend/project/models/User.js 90. backend/project/controllers/user.controller.js 91. backend/project/routes/user.routes.js 92. backend/project/validators/user.validator.js 93. frontend/src/api/users.js

Core pages  94. frontend/src/pages/Home.jsx 95. frontend/src/pages/Lobby.jsx 96. frontend/src/pages/CreateGame.jsx 97. frontend/src/pages/User.jsx 98. frontend/src/pages/Login.jsx 99. frontend/src/pages/Register.jsx 100. frontend/src/pages/ForgotPassword.jsx

Header / appearance / nav  101. frontend/src/components/Header.jsx 102. frontend/src/components/Navbar.jsx 103. frontend/src/components/Greeting.jsx 104. frontend/src/components/Appearance.jsx 105. frontend/src/contexts/AppearanceContext.jsx

Reusable UI components  106. frontend/src/components/Button.jsx 107. frontend/src/components/FormField.jsx 108. frontend/src/components/Spinner.jsx 109. frontend/src/components/Hero.jsx 110. frontend/src/components/GameCard.jsx 111. frontend/src/components/Layout.jsx 112. frontend/src/components/Footer.jsx 113. frontend/src/components/Copyright.jsx 114. frontend/src/components/ProfileImage.jsx 115. frontend/src/components/ConfirmDialog.jsx 116.

frontend/src/components/GameVariantBadge.jsx 117.
frontend/src/components/GameRulesSelector.jsx 118.
frontend/src/components/RoundsSelector.jsx 119.
frontend/src/components/TimeControlSelector.jsx

● TIER 4 — Just know it exists (don't study)

Static content pages 🌱 - frontend/src/pages/AboutUs.jsx - frontend/src/pages/AboutGame.jsx
- frontend/src/pages/Privacy.jsx - frontend/src/pages/Terms.jsx

Seed scripts + data NEW (know npm run seed exists and the order it runs) -
backend/project/seed/db.seed.js - backend/project/seed/users/user.seed.js ·
users/users.json - backend/project/seed/gameCategories/gameCategory.seed.js ·
gameCategories/gamecategories.json - backend/project/seed/trophies/trophy.seed.js -
backend/project/seed/tournaments/tournament.seed.js -
backend/project/seed/matches/match.seed.js -
backend/project/seed/comments/comment.seed.js

Build / config / tooling - backend/project/package.json · package-lock.json · .env -
frontend/package.json · package-lock.json - frontend/vite.config.js · eslint.config.js ·
index.html - backend/.gitignore · frontend/.gitignore · ../.gitignore

Styles (SCSS) — know Sass is the preprocessor; don't read line-by-line -
frontend/src/styles/main.scss · _global.scss · _reset.scss · _variables.scss · main.css ·
main.css.map - Component partials: _Appearance.scss, _BanModal.scss, _Button.scss,
_CommentForm.scss, _CommentItem.scss, _CommentList.scss, _ConfirmDialog.scss,
_Footer.scss, _FormField.scss, _GameCard.scss, _Greeting.scss, _Header.scss, _Hero.scss,
_Navbar.scss, _PlayerInfo.scss, _TournamentCard.scss - Page partials: _Error404.scss,
_Game.scss, _Home.scss, _Lobby.scss, _Login.scss, _Tournament.scss,
_TournamentPage.scss, _User.scss, admin/_AdminLayout.scss, admin/_TournamentCreate.scss

Assets — images + sounds - frontend/src/assets/logo.png - frontend/public/: autumn-
trophy.png, spring-trophy.png, summer-trophy.png, winter-trophy.png, valentines-
trophy.png, default-img.jpg, home-hero.webp, lobby-hero.webp, rules-hero.webp,
tournament-hero.webp, icon.png, team.png, spinner.gif - frontend/public/sounds/: coffee
time.wav, dbl-click.mp3, die-hold.mp3, die-roll.wav, finish-level-sfx.mp3, pling.mp3

REST test scripts (handy for demoing endpoints, not study material) - backend/REST scripts/:
activity.http, comments.http, gamecategories.http, leaderboard.http, matches.http,
queue.http, scripts.md, tournaments.http, trophies.http, user.http

Docs / meta - README.md · README.pdf · NOTES.md · instructions.md - EXAM_PREP.md ·
EXAM_PREP.docx · EXAM_PREP.pdf · FILES_BY_PRIORITY.md (this file)